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Wireless power transfer to a door of a refrigerator appliance

Abstract

A refrigerator appliance having a cabinet defining a chamber and a rotatably hinged door to at least partially seal the chamber or provide access to the chamber. The cabinet includes a resonant LC circuit transmitter and the door includes a resonant LC receiver to transfer electric power from the cabinet to the door.

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Background/Summary

FIELD OF THE INVENTION

(1) The present disclosure is related generally to refrigerator appliances and more particularly to providing electric power to a refrigerator appliance door.

BACKGROUND OF THE INVENTION

(2) Refrigerator appliances generally include a cabinet with one or more chilled compartments, e.g., a fresh food compartment, a freezer compartment, or the like, to maintain foods at low temperatures (i.e., lower than ambient). The chilled compartment(s) of a refrigerator are typically accessible through an opening, with access provided by one or more doors connected by hinges to the appliance cabinet.

(3) Refrigerator appliances having two rotatably mounted opposing doors for access to a single opening, e.g., the fresh food compartment, are generally referred to as “French doors” refrigerators. French doors have desirable features, for example, lighter weight for each door and increase accessibility to the refrigerator cabinet.

(4) However, French doors require additional sealing areas; in particular, the middle portion of the refrigerator opening where the two doors meet must maintain a seal when the doors are closed. Accordingly, some French door refrigerators include an articulating mullion rotatably attached via pivot points or hinges to one of the doors such that access to the compartment via the opening is

not obstructed by the mullion when the door to which the articulating mullion is attached is opened. When closed, each of the doors sealingly engages the mullion with opposing edges of the doors spaced apart for clearance.

(5) Some mullions for French door refrigerator appliances, and in particular articulated mullions, are at least partially formed of thermally conductive materials, such as e.g., metal. Thermally conductive materials are chosen because they typically have advantageous magnetic properties which can facilitate sealing of the doors with a magnetized sealing element when the doors of the refrigerator appliance are in a closed position. However, an inward facing portion of the thermally conductive material is in contact with the chilled air in the compartment and cools an exterior or front wall of the mullion that is in contact with relatively warm ambient air. When the warm ambient air contacts the cool front wall, the warm air is cooled and may cause condensation or “sweat” on the front wall depending on the humidity of the ambient air. The condensation is unsightly and may collect in areas that can cause a safety concern, such as development of mold or mildew, or affect the performance of the refrigerator appliance.

(6) To prevent condensation, some articulating mullions include an electrically powered heating device within the mullion to remedy this undesirable effect. To provide electric power to these heating devices, electrically conductive wires are typically routed from the refrigerator cabinet to the door to which the articulating mullion is attached, and then to the heating element within the mullion. However, the design of some pivot points linking a door to a refrigerator appliance cabinet make the routing of wires undesirable, impractical, or impossible.

(7) Accordingly, refrigerator appliances with improved electric power transfer from the refrigerator cabinet to the refrigerator door to address one or more of the above-described challenges would be beneficial.

BRIEF DESCRIPTION OF THE INVENTION

(8) The present subject matter is directed to a refrigerator appliance having a cabinet and a pivotally attached door that includes features for improved electric power transfer to the door. In particular, the refrigerator appliance includes features that may transfer power from the cabinet to the door without routing electrically conductive wires from the cabinet to the door. Additional aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

(9) In an exemplary aspect, a refrigerator appliance defines a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular. The refrigerator appliance comprises a cabinet comprising a first vertical wall and a second vertical wall spaced laterally from the first vertical wall, a bottom wall and a top wall spaced vertically above the bottom wall, the first and second vertical walls and the bottom and top walls defining a vertical and a lateral dimension of a chamber. The refrigerator appliance further comprises a resonant LC circuit transmitter, and a door rotatably hinged to the cabinet and supported for rotation between a closed position in which an inner surface of the door abuts at least one of the first and second vertical walls and the bottom and top walls to seal a portion of the chamber, and an open position providing access to a portion of the chamber, the door comprising a resonant LC circuit receiver.

(10) In another exemplary aspect, a door assembly for a refrigerator appliance comprises a door comprising an inner surface, an upper portion, a lower portion, and an LC circuit receiver comprising an LC receiver, a bridge rectifier, a buck converter, and an articulating mullion rotatably hinged to the door, the articulating mullion comprising a resistive heating element electrically connected to the buck converter.

(11) These and other features, aspects, and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate

embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
- (2) FIG. 1 provides a front view of a refrigerator appliance according to an exemplary embodiment of the present disclosure;
- (3) FIG. 2 provides a front view of the refrigerator appliance of FIG. 1 with refrigerator doors shown in an open configuration;
- (4) FIG. 3 provides a functional block diagram of a transmitter and a receiver circuit according to an exemplary embodiment of the present disclosure;
- (5) FIG. 4 provides a perspective view of a door and an articulating mullion connected to the door of the refrigerator appliance of FIG. 1; and
- (6) FIG. 5 is a cross-sectional view of doors of an exemplary refrigerator appliance in a closed position and contacting an exemplary articulating mullion according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

- (7) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not a limitation of the invention. It will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.
- (8) Using the teachings disclosed herein, one of skill in the art will understand that the present technology can be used with other types of refrigerators (e.g., side-by-side) or a freezer appliance as well. Consequently, the description set forth herein is for illustrative purposes only and is not intended to limit the technology in any aspect.
- (9) For ease of illustration only, exemplary embodiments are shown on one door, the left door, of a French door refrigerator appliance. One of ordinary skill in the art will understand that the features described can be used on other doors, for example the right door of a French door refrigerator appliance, or on a drawer-type door to a refrigerator or freezer appliance.
- (10) As used herein, the terms “first,” “second,” “third,” and “fourth” may be used to distinguish one component from another and are not intended to signify importance of the individual components. Terms such as “inner” and “outer” refer to relative directions with respect to the interior and exterior of the refrigerator appliance, and in particular the food storage chamber(s) defined therein. For example, “inner” or “inward” refers to the direction towards the interior of the refrigerator appliance. Terms such as “left,” “right,” “front,” “back,” “top,” “bottom,” “above,” or “below” are used with reference to the perspective of a user accessing the refrigerator appliance. For example, a user stands in front of the refrigerator to open the doors and reaches into the food storage chamber(s) to access items therein.
- (11) As used herein, “substantially” means within ten degrees (10°) of the noted direction or within about ten percent (10%) of the noted value or within manufacturing tolerances, whichever margin is greater, unless specifically stated otherwise. Moreover, as used herein, where a wall of

articulating mullion (e.g., front wall) is described as being formed of a particular material, the wall can be considered formed of the particular material even if another material is attached thereto, integrated or embedded into the wall, or coated or plated onto a surface of the wall.

(12) FIG. 1 provides a front view of a refrigerator appliance **100** according to an exemplary embodiment of the present disclosure. Refrigerator appliance **100** extends between a top **101** and a bottom **102** along a vertical direction V. Refrigerator appliance **100** also extends between a first side **105** and a second side **106** along a lateral direction L. Further, refrigerator appliance **100** extends between a front and a back along a transverse direction T (not shown), which is a direction orthogonal to the vertical direction V and the lateral direction L. Vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular and form an orthogonal direction system.

(13) Refrigerator appliance **100** includes a housing or cabinet **120** defining a fresh food chamber **122** and one or more freezer chambers, such as a first freezer chamber **124** and a second freezer chamber **125**, which may both be arranged below fresh food chamber **122** along the vertical direction V. As illustrated, fresh food chamber **122** is bounded by a first vertical wall **108** at the first side **105** and a second vertical wall **110** at the second side **106**, such walls spaced apart in the lateral direction, and horizontal top wall **140** (as a top side) at the top **101** and at the lower boundary by bottom wall **180** (as a bottom side), defining the vertical and lateral dimensions of the chamber **122**. In this configuration, refrigerator appliance **100** may generally be referred to as a bottom mount, or bottom freezer, refrigerator. Cabinet **120** also defines a mechanical compartment (not shown) for receipt of a sealed cooling system (not shown). It will be appreciated that the present subject matter can be used with other types of refrigerator appliances as well, such as e.g., top mount, or top freezer, refrigerator appliances. Consequently, the description set forth herein is not intended to limit the present subject matter in any aspect.

(14) First and second refrigerator doors **126**, **128**, respectively, are hinged and supported for rotation to an edge of cabinet **120** at first **105** and second **106** sides, respectively, between an open position providing access to fresh food chamber **122** as illustrated in FIG. 2 or a closed position in which inner surface **150** of the door abuts the vertical first and second vertical walls **108**, **110** and the top wall **140** and bottom wall **180** of the cabinet, sealing a portion of fresh food chamber **122** as illustrated in FIG. 1. For example, upper and lower hinges **186**, at upper portion **169** and lower portion **159** of exemplary door **126**, may couple each door **126**, **128** to cabinet **120**. When first and second doors **126**, **128** are configured as illustrated in FIGS. 1 and 2, the door arrangement is sometimes referred to as a “French door” configuration. Freezer doors, such as a first freezer door **130** and a second freezer door **131**, may be arranged below refrigerator doors **126**, **128** for accessing one or more freezer chambers, such as first and second freezer chambers **124**, **125**, respectively. In the exemplary embodiment shown in FIG. 1, freezer doors **130**, **131** are coupled to freezer drawers (not shown) slidably coupled within first and second freezer chambers **124**, **125**. Such drawers are thus generally “pull-out” drawers in that they can be manually moved into and out of freezer chambers **124**, **125** on suitable slide mechanisms. Each door **126**, **128**, **130**, **131** can include a handle for accessing one of the chambers **122**, **124**, **125** of refrigerator appliance **100**.

(15) FIG. 2 provides a front perspective view of refrigerator appliance **100** showing refrigerator doors **126**, **128** in an open position to reveal the interior of fresh food chamber **122**.

(16) Door **126** of refrigerator appliance **100** includes an inner surface **150** and an outer surface **152** (FIG. 1). Inner surface **150** generally defines a portion of the interior of fresh food chamber **122** when door **126** is in a closed position as shown in FIG. 1. Outer surface **152** is generally opposite inner surface **150** and defines a portion of the exterior of refrigerator appliance **100** when door **126** is in the closed position. Door **126** includes first and second side surfaces **154**, **155**, respectively, extending between and connecting inner surface **150** and outer surface **152**. As illustrated for example in FIG. 4, the intersection of first inner surface **150** and first side surface **154** form edge **158**. The same construction may result in similarly formed edges at the other intersections of side

surfaces **155**, **156**, **157** and inner surface **151** and outer surfaces **152**, **153**. It will be appreciated that door **128** can be configured in the same or similar manner as door **126** with inner surface **151**, outer surface **153** and third and fourth side surfaces **156**, **157**, respectively, extending between and connecting inner surface **151** and outer surface **153**. Moreover, it will further be appreciated that freezer doors **130**, **131** can likewise include inner, outer, and side surfaces **150**, **152**, **154**.

(17) As further shown in FIG. 2, refrigerator appliance **100** includes various mullions to generally divide the various chambers of refrigerator appliance **100** and/or prevent leakage therefrom. In the present embodiment of a French door refrigerator, refrigerator appliance **100** includes an articulating mullion **200** disposed on first door **126** and a bottom wall **180** disposed between and separating fresh food chamber **122** and first freezer chamber **124**. Refrigerator appliance **100** also includes a stationary mullion **182** disposed between and separating first freezer chamber **124** and second freezer chamber **125**. Bottom wall **180** and stationary mullion **182** generally extend along the lateral direction L between first end **105** and second end **106** of refrigerator appliance **100** and generally extend along the vertical direction V to separate the various chambers of refrigerator appliance **100**. Moreover, although not shown in FIG. 2, bottom wall **180** and stationary mullion **182** generally extend along the transverse direction T approximately the depth of refrigerator appliance **100**.

(18) Refrigerator appliance **100** includes an articulating mullion **200** rotatably coupled or connected to door **126** and positioned so that a long axis of the mullion **200** is parallel to the vertical direction V as shown in FIG. 2. In other embodiments, articulating mullion **200** can be connected to door **128**. In yet other embodiments, articulating mullion **200** can be connected to any suitable door of refrigerator appliance **100**. Moreover, refrigerator appliance **100** can include any suitable number of articulating mullions **200**. For example, where refrigerator appliance **100** has a quad door configuration (i.e., having two rotatably mounted “French door” fresh food doors and two rotatably mounted “French door” freezer doors positioned below the fresh food doors), refrigerator appliance **100** can include one articulating mullion **200** connected to one of the freezer doors and one articulating mullion connected to one of the fresh food doors.

(19) In embodiments of the present application, the refrigerator appliance **100** includes a resonant inductor-capacitor circuit transmitter assembly and a resonant inductor-capacitor circuit receiver assembly, sometimes referred to as, respectively, a resonant LC circuit transmitter assembly and a resonant LC circuit receiver assembly, or simply an LC transmitter and an LC receiver. In the illustrative embodiment shown in FIG. 2, a resonant LC transmitter assembly **160** is shown in bottom wall **180** and a resonant LC receiver assembly **170** is shown in a lower portion **159** of door **126**. The transmitter assembly **160** and the receiver assembly **170** are positioned such that, when the door **126** is closed, the transmitter assembly **160** and the receiver assembly **170** are sufficiently close to each other to allow the transmitter to electronically communicate with the receiver. For example, when the transmitter assembly **160** and the receiver assembly **170** are aligned such that the lateral L dimension and the vertical V dimension of each overlap sufficiently to establish communication from the transmitter assembly **160** to the receiver assembly **170**. Communication can be established when the transmitter assembly **160** produces and emits an electronic signal (e.g., a magnetic field) and the receiver assembly **170** detects the electronic signal. When the transmitter assembly **160** and receiver assembly **170** are in electronic communication with each other, they are sometimes said to be inductively coupled. In the present embodiment, when the door **126** is closed, the transmitter assembly **160** and the receiver assembly **170** are generally proximate to each other and separated by an air gap **171** (FIG. 3) preferably between 0.0 inch (0.0 mm) and 0.5 inch (12.7 mm), more preferably between 0.13 inch (3.3 mm) and 0.38 inch (9.7 mm), more preferably still, the air gap **171** is 0.25 inch (6.4 mm).

(20) In an embodiment, a door assembly **121** comprises exemplary door **126** as described above, an LC receiver circuit assembly, and an articulating mullion. In the exemplary embodiment of the door assembly **121**, the door **126** includes a first inner surface **150** configured to contact the cabinet **120**

when the door is in the closed position of FIG. 1. In particular, the vertical edge of the first vertical wall **108** sealingly contacts the portion of the inner surface **150** at the first side **105** of the cabinet **120**. Similarly, the inner surface **150** at the upper portion **169** of door **126** sealingly contacts the top wall **140** of the cabinet **120** and the inner surface **150** sealingly contacts the bottom wall **180** to seal at least a portion of the fresh food chamber **122** when the door **126** is in the closed position (FIG. 1). Lower portion **159** of door **126** includes a receiver assembly **170** as described below. In an embodiment, the receiver assembly includes an LC receiver **172**, a bridge rectifier **173**, and a buck converter **174** as more thoroughly described below. An articulating mullion **200** is rotatably hinged to door **126** with hinges **186**. In embodiments, the mullion **200** includes a resistive heating element **250** electrically connected, in general, to the receiver assembly **170**, and specifically to the buck converter **174**.

(21) FIG. 3 illustrates transmitter assembly **160** and receiver assembly **170** as a circuit functional block diagram. The transmitter assembly **160** includes a filter stage **161**, an LC tank **162**, a metal-oxide-semiconductor field-effect transistor (MOSFET) **163**, a half bridge inverter **164**, and an interlock circuit **165**. A suitable electric power supply **167** is provided to the transmitter assembly **160** through the cabinet **120** of the refrigerator appliance **100** to provide a prescribed voltage at a prescribed current. The voltage and current supplied by power supply **167** first passes through a filter stage **161** to decouple the supply voltage from the high frequency currents in the LC tank **162**. In some cases, a suitable power supply **167** provides a direct current (DC) voltage (V) to the transmitter assembly **160**. With a DC power supply, the filter stage **161** isolates or decouples the power supplied, for example 12 VDC, from the high frequency current required in the LC tank **162**. In some cases, the LC tank **162** frequency is substantially 125 kHz, although other frequencies may have beneficial results.

(22) The LC tank **162** is an inductor-capacitor circuit, sometimes known as an LC circuit or a resonant circuit. The LC tank is, in general, an electric circuit consisting of an inductor and a capacitor electrically connected in a parallel arrangement. The LC tank **162** can act as an electrical resonator, with electrical current oscillating at the circuit's natural resonant frequency. In some cases, the LC tank **162** is more efficient when operating at its natural frequency. To ensure operation at, or close to, the natural frequency, the LC tank **162** communicates with the MOSFET driver **163**, providing feedback **166** regarding the operating frequency of the LC tank **162**. The MOSFET driver **163** then drives the half-bridge inverter **164** which in turn drives the LC tank **162** at its natural frequency.

(23) In an embodiment of the present disclosure illustrated in FIG. 3, an interlock circuit **165** is placed electronically in-line between the MOSFET driver **163** and the power supply **167**. The interlock circuit **165** selectively allows power to flow from the power supply **167** to the MOSFET driver **163**. When power flows to the MOSFET driver **163**, the MOSFET driver **163** and the half-bridge inverter cooperate to drive the LC tank **162** at its natural resonance frequency as discussed above. When the interlock circuit **165** blocks the flow of power to the MOSFET driver **163**, the LC tank **162** is not driven and current does not oscillate in the LC tank. In an illustrative embodiment, the interlock circuit **165** may include a door sensor **168**, for example a Hall effect sensor, to determine if the door **126** is in a closed position (FIG. 1) or an open position (FIG. 2). In some embodiments, it may be desirable for the interlock circuit **165** to allow power to flow to the MOSFET driver **163** when the door **126** is in the closed position and to prevent the flow of power when the door **126** is not closed.

(24) Beneficially, the oscillating current in the LC tank **162** creates a magnetic field. It is understood that a uniform oscillation frequency, in this case the natural resonant frequency of the LC tank, creates a uniform and consistent magnetic field. The LC tank **162** provides a magnetic field that is coupled with the LC receiver **172** in the receiver assembly **170** located in the lower portion **159** of door **126**. When the magnetic field of the LC tank **162** is coupled with the LC receiver **172**, the transmitter assembly **160** and receiver assembly **170**, in general, and the LC tank

162 and the LC receiver **172** in particular, may be said to be inductively coupled. In an embodiment of the present disclosure, the natural resonant frequency of the LC receiver **172** is substantially the same as the natural resonant frequency of the LC tank **162** (i.e., the natural resonant frequency of the receiver assembly **170** substantially matches the natural resonant frequency of the transmitter assembly **160**). Under some circumstances it may be beneficial for the frequencies of the LC tank **162** and the LC receiver **172** to differ one from the other, with the difference varying by up to, and including, 10%.

(25) The magnetic field generated by the LC tank **162** induces an alternating current in the LC receiver **172** through inductive coupling. Thus, through inductive coupling, electric power is transferred generally from the transmitter assembly **160** to the receiver assembly **170**, and specifically from the LC tank **162** to the LC receiver **172**. The receiver assembly **170** may include a bridge rectifier **173** as shown in the embodiment of FIG. 3. The bridge rectifier is electrically connected to the LC receiver **172** and rectifies, or converts, the alternating current to a direct current. The direct current may be routed using current-carrying wires to electric loads in the door **126**. In some embodiments, it may be desirable to reduce the direct current voltage output of the bridge rectifier **173** with a buck converter **174**. A buck converter is sometimes known as a direct current to direct current (DC-to-DC) step-down converter, and is used to reduce, or “step down,” voltage from an input to an output. The buck converter **174** reduces (i.e., “steps down”) the direct current voltage from the rectifier **173** to a predetermined direct current voltage appropriate for the designed electric loads **178**. The control circuit **176** may direct and selectively apply the voltage to electric loads **178** through current-carrying conductors or wires **177**. In some embodiments, control circuit **176** may include sensors (e.g., thermometer(s) or hygrometer(s)) or logic elements to determine the application of the voltage. The reduced direct current voltage may be routed using current-carrying wires to electric loads **178** in the door **126**, for example a light-providing device such as a light emitting diode (LED) (not shown) or a heating element **250** (described below).

(26) Referring now to FIGS. 4 and 5, FIG. 4 provides a perspective view of an exemplary first door **126**, bottom wall **180**, and articulating mullion **200** connected to door **126** of exemplary refrigerator appliance **100** of FIGS. 1 and 2. FIG. 5 provides a close-up, cross-sectional view of first and second doors **126**, **128** of exemplary refrigerator appliance **100** in a closed position and contacting articulating mullion **200** according to an exemplary embodiment of the present disclosure.

(27) As shown in FIG. 4, articulating mullion **200** can be rotatably coupled or rotatably hinged, via hinges **186**, to door **126**. Articulating mullion **200** can be rotated or articulated about a vertical axis V.sub.1, which extends along the vertical direction V through hinges **186** as shown.

(28) As shown in the illustrative embodiment of FIGS. 4 and 5, articulating mullion **200** includes a body **202** which is shown as a generally rectangular cross-sectional shape for ease of illustration. It will be appreciated that body **202** can have any suitable cross-sectional shape as will be apparent to an ordinarily skilled artisan. Also for ease of illustration, body **202** is shown as generally hollow. In some embodiments, body **202** may be filled with an insulating material, for example polyurethane or expanded polystyrene. Body **202** extends between a top portion **204** and a bottom portion **206** along the vertical direction V, between a first end **208** and a second end **210** along the lateral direction L (FIG. 5), and between a front face **212** and a rear face **214** along the transverse direction T.

(29) Articulating mullion **200** includes a tab **216** extending from body **202** as shown in FIG. 3. For this exemplary embodiment, tab **216** extends from top portion **204** of body **202**. In some embodiments, tab **216** can extend from bottom portion **206** of body **202**. In yet other embodiments, body **202** can include tabs **216** extending from both top portion **204** and bottom portion **206**. Tab **216** may be sized and shaped to fit within and interact with a groove **184** defined in cabinet **120** of refrigerator appliance **100** (FIG. 2). For example, groove **184** may include cam surfaces that may interact with tab **216** to cause rotation of articulating mullion **200** from a first position to a second

position when door **126** is rotated from a closed position (FIGS. **1** and **3**) to an open position (FIG. **2**) or vice versa.

(30) As shown in FIGS. **4** and **5**, body **202** includes a front wall **220** having a front face **212** and a rear face **224** opposite front face **212**. When door **126** is in the closed position FIG. **4**, or when both door **126** and door **128** are in the closed position FIG. **5**, front wall **220** is oriented in a plane parallel to the vertical and lateral directions V, L. Likewise, front face **212** of front wall **220** is coplanar with the vertical and lateral direction V, L. Front face **212** of front wall **220** faces the exterior of refrigerator appliance **100** and rear face **214** of rear wall **226** faces the interior of refrigerator appliance **100** when door **126** is in a closed position.

(31) In the illustrative embodiment of FIG. **5**, within the body **202** is resistive heating element **250**, generally centered along the lateral dimension of rear face **224** of front wall **220**. The resistive heating element **250** may be any type heating element capable of converting electrical power to thermal energy, or heat, through, for example, Joule heating. For example, heating element **250** may be a metallic wire, ribbon, foil, or strip comprising a metallic alloy (e.g., an alloy of nickel and chromium), or a ceramic material, or a semiconductor, having a resistance to the flow of electric current. The heating element **250** may continuously extend along a portion or along the full vertical V length of the mullion **200**, or may be made up of a plurality of segments spaced apart and extending along all or part of vertical V length of the mullion **200**. In an exemplary embodiment of the present disclosure, heating element **250** may selectively receive power from LC receiver assembly **170** of FIG. **3**. More specifically, buck converter **174** may provide a suitable voltage for heating element **250** through control circuit **176**. In an embodiment, the electric load **178** may be the heating element **250**. Power may be supplied to the heating element **250** through wires **177** routed from the receiver assembly **170** in the lower portion of door **126** and through the hinge **186** as illustrated in FIG. **5**.

(32) As shown in FIG. **5**, when doors **126**, **128** are in a closed position, a gap G is defined between doors **126**, **128**. Ambient air **192**, which is generally warm relative to the cooled or chilled air of fresh food chamber **122** (or similarly first or second freezer chambers **124**, **125**) of refrigerator appliance **100**, flows through gap G and contacts front face **212** of front wall **220** of articulating mullion **200**. As articulating mullion **200** is positioned to block the airflow through gap G, articulating mullion **200** prevents relatively warm ambient air **192** from leaking into refrigerator appliance **100**. Articulating mullion **200** also prevents cooled or chilled air from flowing out of refrigerator appliance **100**. To prevent such leakage, first and second inner surfaces **150**, **151** of each door **126**, **128**, respectively, or one or more gaskets **252** along such inner surfaces **150**, **151**, contact front face **212** of articulating mullion **200**. To hermetically seal front face **212** with doors **126**, **128**, each door **126**, **128**, or gaskets **252**, and articulating mullion **200** can include magnets or comprise materials having magnetic properties to seal doors **126**, **128** in sealing engagement with articulating mullion **200**.

(33) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the

refrigerator appliance comprising: a cabinet defining a fresh food chamber and a freezer chamber, the cabinet comprising: a first vertical wall and a second vertical wall spaced laterally from the first vertical wall; a bottom wall and a top wall spaced vertically above the bottom wall, the first and second vertical walls and the bottom and top walls defining a vertical and a lateral dimension of a chamber, the bottom wall being disposed between the fresh food chamber and the freezer chamber; and a resonant LC circuit transmitter disposed in the bottom wall; a door rotatably hinged to the cabinet and supported for rotation between a closed position in which an inner surface of the door abuts at least one of the first and second vertical walls and the bottom and top walls to seal a portion of the chamber, and an open position providing access to a portion of the chamber, the door comprising a resonant LC circuit receiver, the resonant LC circuit receiver comprising a bridge rectifier and a buck converter; and an articulating mullion rotatably hinged to the door, the articulating mullion comprising a resistive heating element electrically connected to the buck converter, wherein the buck converter steps down a direct current from the bridge rectifier to a predetermined direct current voltage, and wherein the resonant LC circuit transmitter and the resonant LC circuit receiver are inductively coupled in the closed position of the door, and wherein a current is induced in the resonant LC circuit receiver when the resonant circuit transmitter and the resonant LC circuit receiver are inductively coupled.

2. The refrigerator appliance of claim 1, wherein the inductive coupling of the resonant LC circuit transmitter and the resonant LC circuit receiver transfers electric power from the resonant LC circuit transmitter to the resonant LC circuit receiver.
3. The refrigerator appliance of claim 1, wherein the resonant LC circuit transmitter comprises: a power supply providing prescribed voltage at a prescribed current; a filter stage; an LC tank; a MOSFET driver; and a half bridge inverter, wherein the MOSFET driver and the half bridge inverter cooperate to drive the LC tank at a frequency to produce a magnetic field.
4. The refrigerator appliance of claim 3, wherein the MOSFET driver and the half bridge inverter are in electronic communication with the LC tank to drive the LC tank at a resonant frequency.
5. The refrigerator appliance of claim 4, wherein the LC tank provides feedback to the MOSFET driver.
6. The refrigerator appliance of claim 3, wherein the transmitter includes an interlock circuit placed electrically in-line between the MOSFET driver and the power supply to selectively allow power to flow from the power supply to the MOSFET driver.
7. The refrigerator appliance of claim 6, wherein the interlock circuit includes a door sensor to determine when the door is in the closed position.
8. The refrigerator appliance of claim 7, wherein the door sensor is a Hall effect sensor.
9. The refrigerator appliance of claim 6, wherein the power supply provides electrical power when the door is in the closed position.
10. The refrigerator appliance of claim 3, wherein the filter stage decouples a high frequency current in the LC tank from the power supply.
11. The refrigerator appliance of claim 1, wherein a natural resonance frequency of the resonant LC circuit receiver matches a natural resonance frequency of the LC tank.
12. The refrigerator appliance of claim 1, wherein the current induced in the resonant LC circuit receiver is an alternating current and is converted to a direct current by the bridge rectifier.
13. The refrigerator appliance of claim 1, wherein the resonant LC circuit transmitter and the resonant LC circuit receiver are inductively coupled across an air gap.
14. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, the refrigerator appliance comprising: a cabinet comprising: a first wall, a second wall, a top wall, and a bottom wall, the first wall and the second wall being spaced apart along the lateral direction, the top wall and the bottom wall being spaced apart along the vertical direction, the first wall, the second wall, the top wall, and the bottom wall defining a chilled chamber, a resonant LC circuit transmitter mounted on a front face of the bottom wall, the resonant LC circuit transmitter

configured to generate an electronic signal, and a door rotatably hinged to the cabinet, the door comprising: a resonant LC circuit receiver being configured to receive the electronic signal generated from the resonant LC circuit transmitter when the door is in a closed position thereof, the resonant LC circuit receiver comprising a LC receiver, a bridge rectifier, and a buck converter, and an articulating mullion rotatably hinged to the door, the articulating mullion comprising a resistive heating element electrically connected to the buck converter, wherein the resonant LC circuit transmitter and the resonant LC circuit receiver are inductively coupled in the closed position of the door to transfer power to the heating element within the articulating mullion.
